CS341 · COMPUTER GRAPHICS

# Mock Midterm Solutions

20 marks · 1 hour · same format as the past midterms

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## 01 Solution Q1

1. The **frame buffer** (refresh buffer, bit map) is the memory area that stores the picture of a raster system; each location corresponds to one pixel.
2. **Persistence** is the time for the emitted light to decay to one-tenth of its original intensity. **Low** persistence is better for animation.

3) $v = Q - P = (1 - 4,\ 5 - 1) = (-3,\ 4)$ $\quad |v| = \sqrt{(9 + 16)} = 5$

4) $$AR = \frac{\frac{720}{6}}{\frac{1280}{10}} = \frac{120}{128} = 0.9375$$

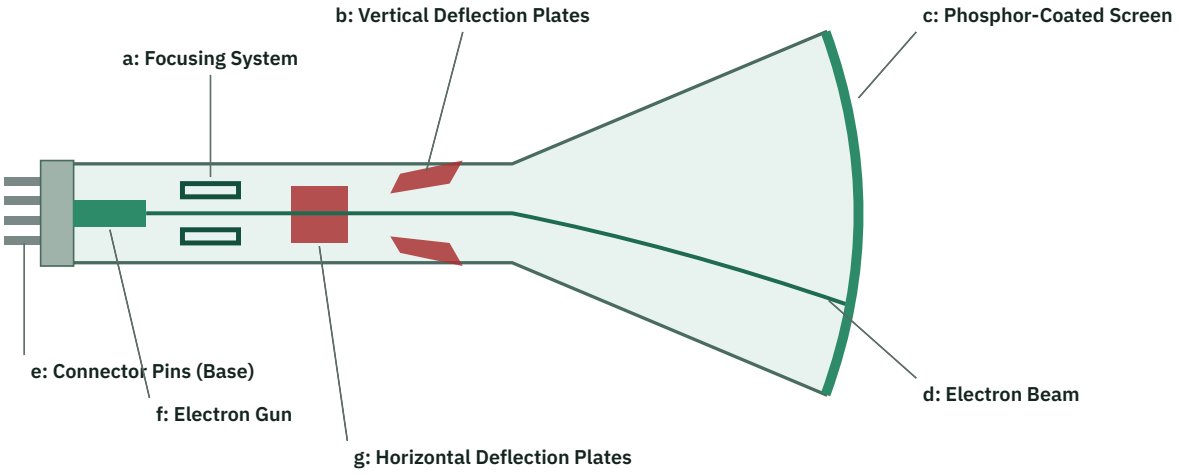
## 02 Solution Q2

| # | Answer | Reason |
|---|---|---|
| 1 | **True** | |
| 2 | **False** | Plasma is emissive; LCD is the nonemissive one. |
| 3 | **True** | |
| 4 | **True** | |
| 5 | **True** | |
| 6 | **True** | And x increases by $\frac{1}{m}$. |
| 7 | **False** | To one-tenth, not half. |
| 8 | **False** | The result is a scalar. |

## 03 Solution Q3

1. b  2. c  3. b  4. c  5. b  6. b  7. b  8. b

## 04 Solution Q4




a: Focusing System  b: Vertical Deflection Plates  c: Phosphor-Coated Screen  d: Electron Beam

e: Connector Pins (Base)  f: Electron Gun  g: Horizontal Deflection Plates

## 05 Solution Q5

$$dx = 4, \quad dy = 8, \quad m = \frac{8}{4} = 2 > 1$$

$$steps = max(|dx|, |dy|) = 8$$

$$xInc = \frac{4}{8} = 0.5 \qquad yInc = \frac{8}{8} = 1$$

$$round(a) = int(a + 0.5)$$

| k | x | y | round(x) | round(y) |
|---|---|---|---|---|
| 0 | 1 | 2 | 1 | 2 |
| 1 | 1.5 | 3 | 2 | 3 |
| 2 | 2 | 4 | 2 | 4 |
| 3 | 2.5 | 5 | 3 | 5 |
| 4 | 3 | 6 | 3 | 6 |
| 5 | 3.5 | 7 | 4 | 7 |
| 6 | 4 | 8 | 4 | 8 |
| 7 | 4.5 | 9 | 5 | 9 |
| 8 | 5 | 10 | 5 | 10 |

